Aaron Kakembo

Full-Stack Engineer | Fintech Systems | AI Engineer

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PROFESSIONAL SUMMARY

I am fascinated by greatness and I want to be one of the greats. I look up to Linus Torvalds, Elon Musk, Steve Jobs and Satoshi Nakamoto, and I would like to have the same impact on the tech world. That is why I am always studying something new on the side, and I play some chess.

Since 2021 I have been building and shipping software, mostly payment systems and AI tools, from the database to the screen to deployment. Through the Refactory Apprenticeship Program I also led a team of five as technical lead on SplitFair.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, Rust, SQL (PL/pgSQL), Bash, PowerShell, HTML/CSS
Backend & APIs:	FastAPI, Flask, Node.js, Rust (Axum, Tokio, SQLx), REST, webhooks, Server-Sent Events
Frontend:	React, Next.js, React Native / Expo, Vite, Tailwind CSS, React Three Fiber
Fintech:	MTN and Airtel Mobile Money, idempotency keys, reconciliation and retry logic, integer money math, double-entry ledgers, payment state machines
AI & Agents:	LangGraph, CrewAI, OpenAI Agents SDK, MCP, RAG with FAISS, pgvector, Whisper, Vercel AI SDK, Gradio
Data & Infra:	PostgreSQL (RLS, RPCs), Supabase, SQLite, SQLAlchemy, Alembic, Docker, Fly.io, Vercel, Hugging Face Spaces, Nginx, Caddy, Linux VPS, GitHub Actions
Testing:	Vitest, pytest, Jest, Playwright, React Testing Library, pgTAP

PROFESSIONAL EXPERIENCE

Software Engineer 2021 – Present

Product builds: payments, marketplaces and AI tools · Kampala, Uganda · Remote

- **Swap Africa** (swapafrica.online): buying and selling online between strangers is risky for both sides. Swap holds the buyer's money safely, a rider delivers and photographs the item, and the seller is paid only when the buyer confirms receipt with a QR code. I built the whole platform: the system that handles the money, the web apps for sellers, buyers and admins, and the rider's phone app. Rust, Next.js, React Native
- **broRacks** (broracks.online): businesses that want to take mobile-money payments normally have to connect to several providers. broRacks lets them collect money from customers, and pay money out to phones and bank accounts, through one connection. I built it end to end, including merchant sign-up, self-service access keys, payment notifications and the deployment. Python, FastAPI, PostgreSQL
- **tvPay** (tvpay.app): an online shop where customers pay for satellite-TV subscriptions from their phone. I built it with safeguards so a customer can never be charged the wrong price or twice, every order tracked from payment to delivery, an admin dashboard, and automated tests on the payment paths. Next.js, Supabase, Docker
- **JijiLens** (jijilens.fly.dev): finding the product you saw in a photo is hard on a classifieds site. JijiLens lets you upload a photo, tap one item, and see similar listings from Jiji Uganda. I built it to collect public listings politely, without storing seller names or phone numbers, to match photos by how they look, and to run as a live demo. Python, FastAPI, PostgreSQL
- **Sales agent:** finding customers one by one is slow for a startup. This AI agent finds small businesses, starts a WhatsApp conversation with each, answers their replies and passes interested people to me. I built it with strict safety limits, such as never messaging the same person twice and stopping itself when something looks wrong. Python, Node.js

SplitFair May – Sep 2026

Technical Lead · Refactory Apprenticeship Program team project, team of five · splitfair.africa

- A platform where brands and agencies hire creators, fund the contract up front, and the payment is split fairly between everyone who worked on it. It solves the "who gets paid what, and when" problem for creative work.
- I led a team of five and made the main technical decisions. I built the core system that holds, tracks and pays out the money, and led the background services that process media, deliver files and run checks. I also contributed to the web and mobile apps.

Electoral Commission, General Elections Jan – Feb 2026

Presiding Officer · Uganda

- Ran a polling station and led its team of polling assistants through election day. When the fingerprint voter-check machines failed, materials arrived late and riots broke out, I switched to the paper register, kept order and protected the secrecy of the vote. After closing I counted in front of party agents and the public, recorded the results on the official form and handed over every ballot.

SELECTED PROJECTS

Mimic, video-to-code agents Python | LangGraph | Gemini | Open Source

github.com/aaron-official/mimic

- Rebuilding an app by following a coding tutorial is slow. Mimic does it for you: AI agents watch the video, write the app and test it, and a supervisor steps in when one gets stuck. I designed and wrote the whole system.

Hybrid Meeting Minutes Generator Whisper | Gradio | HF Spaces | 3 GitHub stars

huggingface.co/spaces/aaron-official/hybrid-meeting-minutes-generator

- Nobody likes typing up meeting notes. This tool turns a recording of a meeting into written minutes in 7 languages. I built and deployed it, and it is live on Hugging Face and starred by 3 people on GitHub.

ocs4dev, fintech API assistant Python | FAISS | Gradio | HF Spaces

huggingface.co/spaces/aaron-official/ocs4dev

- Developers lose hours searching long, confusing documentation. This assistant answers questions about connecting to payment services, using the documentation itself as its source. I built it and put it live on Hugging Face.

Advanced Image Processing Suite Python | Gradio | OpenCV | HF Spaces

huggingface.co/spaces/aaron-official/Advanced-Image-Processing-Suite

- Editing images normally means juggling several separate tools. This web app brings them into one simple page: it converts, enhances and edits images, removes backgrounds, generates new images with AI and handles many images at once. I built it, with automated tests, and it is live on Hugging Face.

strom, audiobook converter Rust | Tokio | Open Source

github.com/aaron-official/strom

- Many players cannot open the original audiobook files, and converting them is slow. This tool converts audiobooks to a format more devices can play, several files at once and with the chapters kept. I built it and it is open source.

EDUCATION & TRAINING

Refactory Apprenticeship Program 2026

Kampala, Uganda · Six-month intensive software engineering apprenticeship (refactory.academy)

- **Learning phase (8 weeks):** UML and system modelling, object-oriented design, test-driven development (red-green-refactor, mocks and stubs, working with legacy code), behaviour-driven development with Gherkin, SOLID principles, design patterns and React. My exercises are in this repository.
- **Project phase (17 weeks):** collaborative, sprint-based development of a real product. I was technical lead of a team of five on SplitFair (see above), responsible for the architecture, the Rust API and the worker platform.

Self-directed learning: Udemy, Zero To Mastery and YouTube 2018 – Present

Online courses and tutorials

- I started learning to code in 2018 through Udemy courses, Zero To Mastery and YouTube tutorials, and I still study something new on the side. That is where the skills I now use to build and ship software began.

LICENSES & CERTIFICATIONS

Certification for Software Engineer

HackerRank · Oct 2026 · Credential ID 88C94623B65C · Show credential

- Role certification from HackerRank's timed skills assessment, covering problem solving, SQL and REST APIs.

REST API Certificate

HackerRank · Oct 2026 · Credential ID 79A5B5D88550 · Show credential

- Skills certification on RESTful principles, HTTP methods, authentication and API design, the same ground I use daily in broRacks, tvPay and the Swap API.